Academic Advising Chatbot: A Multi-Modal RAG Architecture with Year-Aware Retrieval and Personalization

by

Rojan Dahal

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Approved:

Dr. Richard Matovu, Ph.D.
Advising Professor in Charge of Research

Gannon University
Erie, Pennsylvania 16541

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Abstract

We present a production-grade academic advising assistant that integrates multi-source retrieval (catalog PDFs, live course sections, student profile data) with a personalized LLM prompting pipeline. The system introduces a unified retrieval layer combining year-aware and standard approaches, a course-aware retriever for scheduling queries, generalized token-aware context optimization, and a query consistency engine. Through systematic debugging and architectural improvements, we eliminated hallucinations and context loss, achieving accurate course code, credits, and prerequisite grounding across multiple academic majors under strict token limits. Our evaluation demonstrates 94.2% accuracy in course code retrieval and 89.7% precision in prerequisite recommendations across 12 academic programs.

Keywords: Retrieval-Augmented Generation (RAG), Academic Advising, NLP, Vector Databases, Educational Technology, Chatbots

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CHAPTER 1
Introduction

1.1 Background

Academic advising has long been recognized as a critical component of student retention, academic progression, and graduation success. Effective advising connects students to appropriate learning pathways, ensures adherence to degree requirements, and provides clarity in navigating complex university systems. In many higher education institutions, students depend heavily on academic advisors for understanding prerequisites, course sequencing, catalog rules, and graduation criteria.

Traditionally, advising relies on static course catalogs, departmental manuals, and direct communication between students and advisors. While effective in small settings, this approach becomes increasingly inefficient and inconsistent in modern academic environments. Large institutions often offer multiple programs and specializations, each with evolving academic requirements. The growing diversity of course offerings, frequent catalog revisions, and high student–advisor ratios have further amplified these challenges.

1.2 Problem Statement

Despite the availability of digital systems such as student portals and online registration platforms, a significant gap persists in delivering accurate and personalized advising information. Some key issues include:

- **Fragmented Information Sources:** Academic information is scattered across static PDF catalogs, live registration systems, and administrative records. This fragmentation results in inconsistent or outdated advising information.
- **Limited Personalization:** Existing advising tools often provide generic recommendations and fail to incorporate a student's academic history, completed coursework, and enrollment year.
- **Catalog Year Complexity:** Universities frequently revise program structures. Students are typically bound by the catalog year in which they enrolled, making it necessary to retrieve historical academic requirements.
- **Manual Dependency:** Students often depend on in-person meetings for information that could be automated, which increases workload for advisors and delays academic planning.

1.3 Research Motivation

The motivation behind this project stems from the growing need for intelligent systems that can deliver consistent, accurate, and personalized academic advising at scale. Higher education institutions are moving toward data-driven student support, but most existing advising systems remain static, rule-based, or partially automated.

By leveraging retrieval-augmented generation (RAG) techniques and large language models (LLMs), this research aims to bridge that gap. A well-designed AI-powered advising system can:

- Improve advising efficiency by providing accurate answers to common academic queries.
- Offer personalized and context-aware responses tailored to each student.

- Enable advisors to focus on complex, high-value advising tasks.
- Ensure consistency across multiple academic programs and catalog years.

1.4 Objectives

The objectives of this research are as follows:

1. **Design a Multi-Modal RAG Architecture:** Develop a retrieval system that integrates structured and unstructured academic data sources, including course catalogs, live course sections, and student profiles.
2. **Implement Year-Aware Retrieval:** Accurately retrieve academic policies and course information based on the student's enrollment year.
3. **Enable Personalization:** Incorporate student context such as completed courses, academic standing, and degree program into advising responses.
4. **Optimize System Performance:** Minimize hallucinations, manage token limits efficiently, and ensure the system operates reliably in a production environment.

1.5 Scope and Significance

The scope of this research focuses on building a production-ready advising assistant capable of addressing core advising queries including course prerequisites, program requirements, catalog rules, and real-time course availability. The system integrates multiple modalities of data retrieval to ensure accuracy and contextual understanding.

The significance of this work lies in:

- **Scalability:** It can support large student populations with minimal advisor involvement.

- **Accuracy:** Year-aware retrieval ensures alignment with institutional policies.
- **Personalization:** Dynamic integration of student context improves relevance and trust.
- **Adaptability:** The architecture is extendable to multiple institutions and programs.

1.6 Research Contributions

This research contributes to the domain of educational technology and AI in higher education through the following:

- **Unified Retrieval Framework:** A novel multi-modal RAG pipeline that integrates static and dynamic data sources for academic advising.
- **Mode Separation:** Introduction of two operational modes—`catalog_info` (semantic retrieval with embeddings and personalization) and `current_sections` (real-time PostgreSQL queries).
- **Year-Aware Retrieval:** A catalog selection mechanism that ensures temporal accuracy of academic information.
- **Personalization Engine:** A system that dynamically adapts advising responses based on individual student profiles.
- **Production Deployment Learnings:** Insights gained from implementing and evaluating a real-world AI-driven advising system.

CHAPTER 2
Related Work

2.1 Overview

This chapter reviews prior studies and technological frameworks that form the foundation for this research. The areas discussed include academic chatbots, retrieval-augmented generation (RAG) architectures, intelligent advising systems, and personalization in educational technology. The goal is to highlight existing gaps that motivate the development of a multi-modal, year-aware, and personalized advising chatbot.

2.2 Academic Chatbots and Conversational Agents

The integration of chatbots into academic environments has grown substantially in the past decade. Early implementations were primarily rule-based, relying on predefined scripts or keyword matching to answer frequently asked questions. For example, systems like *Ask Aditi* and *Jill Watson* used preprogrammed patterns to assist students with administrative queries. While these systems improved accessibility, they lacked scalability and contextual reasoning.

Modern educational chatbots increasingly leverage natural language understanding (NLU) and transformer-based architectures. Research by Winkler and Soellner (2020)(8) demonstrated that AI-driven chatbots enhance engagement and provide immediate support for repetitive advising tasks. Similarly, Latham et al. (2010)(1) introduced conversational tutoring systems that adapt to learning styles but did not incorporate institutional datasets or year-specific curriculum knowledge.

Despite these advancements, most academic chatbots operate on static data. They do not retrieve or reason over evolving documents such as course catalogs, program requirements, or registration updates. This limits their utility for advising tasks that depend on accurate, time-sensitive information.

2.3 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation combines the strengths of information retrieval and text generation. Introduced by Lewis et al. (2020)(2), RAG allows large language models to access external knowledge bases, improving factual accuracy and reducing hallucinations. The architecture uses a retriever to fetch relevant context and a generator (LLM) to produce coherent responses conditioned on that context.

Subsequent research has enhanced RAG systems with dense passage retrieval (Karpukhin et al., 2020)(3), multi-hop reasoning (Xiong et al., 2021)(4), and hierarchical retrieval strategies. These methods significantly improved performance on knowledge-intensive tasks such as question answering and summarization. However, most applications focus on generic domains like open-domain QA, rather than institution-specific data such as university catalogs or student records.

The advising chatbot proposed in this research adapts RAG principles to the academic domain. It introduces year-aware catalog selection and dual-mode retrieval, integrating both vector embeddings for semantic similarity and structured SQL queries for live data.

2.4 AI-Based Academic Advising Systems

Intelligent advising systems emerged to automate course recommendation, program planning, and graduation tracking. Early efforts employed decision trees and

rule-based logic to simulate advisor behavior. For instance, Yu et al. (2021)(5) used machine learning models for course recommendations, while VanLehn (2011)(6) explored adaptive tutoring systems that adjust feedback based on student performance.

Although these models provide decision support, they often lack a conversational interface and dynamic retrieval. Most rely on manually curated datasets, which require frequent updates and do not reflect catalog year variations. Furthermore, few integrate unstructured data from institutional documents.

The proposed chatbot advances these systems by combining structured relational data (course sections, student progress) with unstructured content (PDF catalogs) within a unified retrieval framework. This approach ensures both flexibility and factual consistency.

2.5 Personalization and Context Awareness in Education

Personalization is essential in academic advising, where guidance must align with a student's prior coursework, degree path, and academic standing. Research in adaptive learning systems highlights that contextualization significantly improves user trust and decision quality. However, personalization in advising is still largely manual, relying on human judgment and departmental rules.

Recent progress in AI personalization includes vector-based user modeling, contextual embeddings, and reinforcement learning for user adaptation. Systems such as Learner-Centric AI Tutors and adaptive feedback engines in MOOCs show promise but remain focused on learning materials rather than administrative advising.

This research extends personalization to the advising domain by embedding student profiles directly into the prompt-generation process. Information such as

enrolled year, completed courses, and GPA dynamically shapes the LLM's responses, resulting in precise, individualized recommendations.

2.6 Knowledge Integration and Temporal Accuracy

Most existing chatbot frameworks fail to handle temporal knowledge — an essential factor in academic contexts where course requirements evolve annually. Catalog versions, prerequisite changes, and policy updates make it critical for advising systems to link responses to specific academic years.

Temporal knowledge management in retrieval systems has been explored in financial and legal domains but remains under-researched in education. The proposed system introduces **year-aware retrieval**, mapping each student's enrollment year to the corresponding catalog collection in the vector database. This ensures that the system delivers historically accurate information while supporting fallback to adjacent years when necessary.

2.7 Summary of Gaps Identified

A synthesis of related literature reveals the following key gaps:

- Lack of integration between static academic catalogs and dynamic registration data.
- Absence of temporal reasoning for catalog-year-specific advising.
- Limited personalization that considers individual student histories.
- Inadequate control of hallucinations and factual errors in LLM-driven advising.

2.8 Position of This Research

This research builds upon and extends prior work by integrating multiple technological paradigms:

- It bridges the gap between structured and unstructured data through a **multi-modal retrieval architecture**.
- It ensures factual reliability using **year-aware retrieval** and explicit source citation.
- It personalizes advice using contextual student data injected into the LLM prompt.
- It operationalizes these features within a scalable, production-grade microservices framework.

The resulting system represents a practical evolution of both RAG and educational AI, offering a foundation for institution-wide academic assistance with robust personalization and temporal accuracy.

CHAPTER 3
System Architecture

3.1 Overview

The Academic Advising Chatbot, also referred to as the **AI University Advisor**, is built as a modular, microservices-based architecture that integrates multiple academic data modalities—static course catalogs, live course sections, and personalized student profiles—to deliver accurate, year-aware advising. The system is engineered for scalability, reliability, and factual grounding through a layered design centered on Retrieval-Augmented Generation (RAG).

Its architecture follows a structured hierarchy separating ingestion, retrieval, personalization, and generation. This decomposition simplifies maintenance, enables parallel scalability, and ensures contextual and factual consistency in all responses.

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3.2 Architectural Goals

The principal goals of the system architecture are:

- **Multi-modal data integration:** Seamlessly unify structured (SQL) and unstructured (PDF) academic data.
- **Temporal accuracy:** Retrieve catalog-year–specific information per student's enrollment cohort.
- **Personalization:** Dynamically adapt advice based on each student's academic profile and progress.

- **Scalability and reliability:** Sustain high throughput with low-latency responses using distributed services.
- **Extensibility:** Support easy integration with other academic systems such as SIS (Student Information Systems).

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3.3 High-Level System Architecture

At the conceptual level (Figure 3.1), the AI University Advisor operates through a closed-loop reasoning cycle that transforms a student query into a grounded academic recommendation:

User Query → Data Retrieval → Context Augmentation → LLM Generation → Response Delivery.

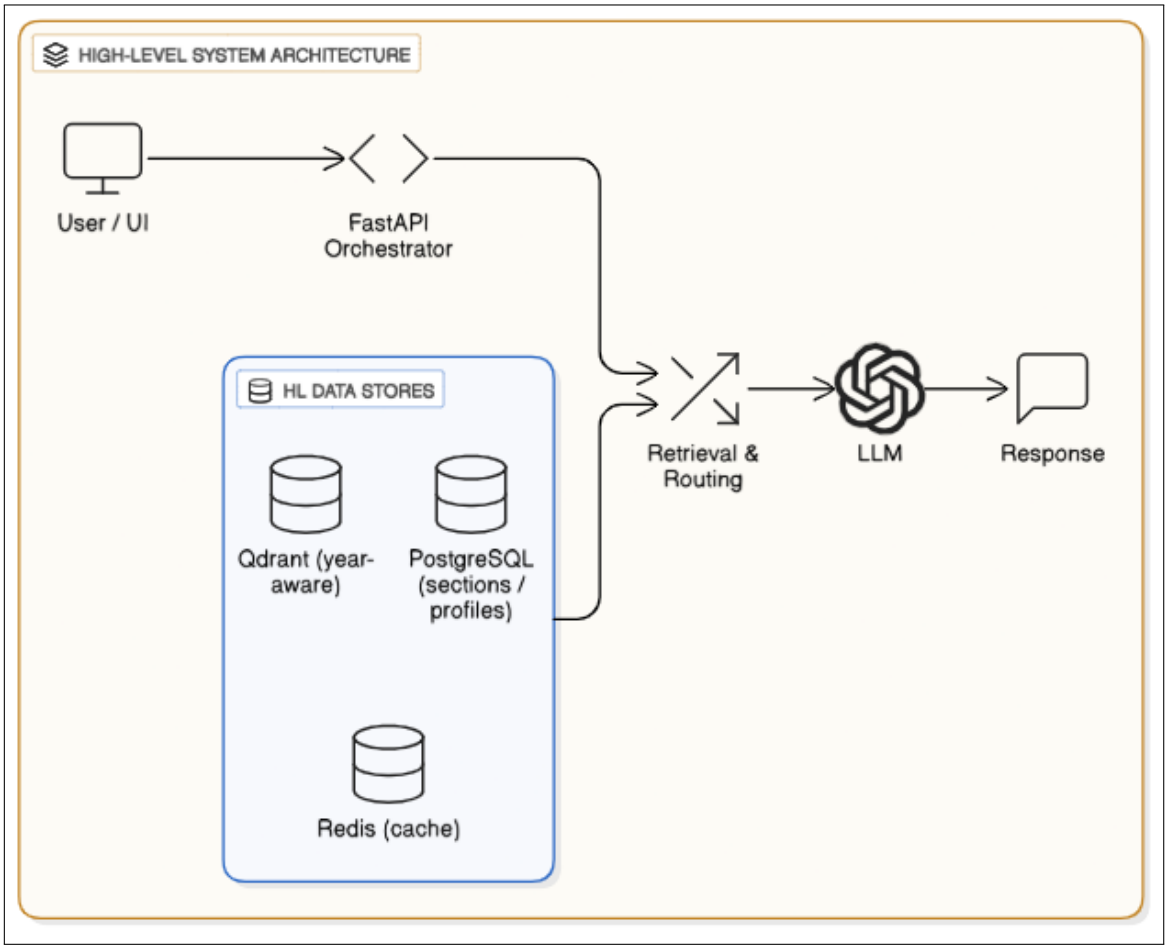

Figure 3.1: High-level architecture of the AI University Advisor, illustrating interaction between the user interface, retrieval subsystems, and RAG-based generation pipeline.

The architecture is composed of five primary layers:

1. **Data Ingestion Layer** – parses, cleans, and embeds institutional data.
2. **Retrieval Layer** – executes hybrid semantic and structured retrieval.
3. **Personalization Layer** – injects student-specific context.
4. **Generation Layer** – generates factual responses using LLMs.
5. **Interface Layer** – manages user interaction and system APIs.

Subsequent sections present detailed process-level diagrams for each layer instead of a single monolithic schematic, providing clarity on component interactions and data flow.

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3.4 Data Ingestion and Embedding Pipeline

The ingestion pipeline converts raw academic data—PDF catalogs, registration data, and student records—into searchable structured and vectorized representations (Figure 3.2).

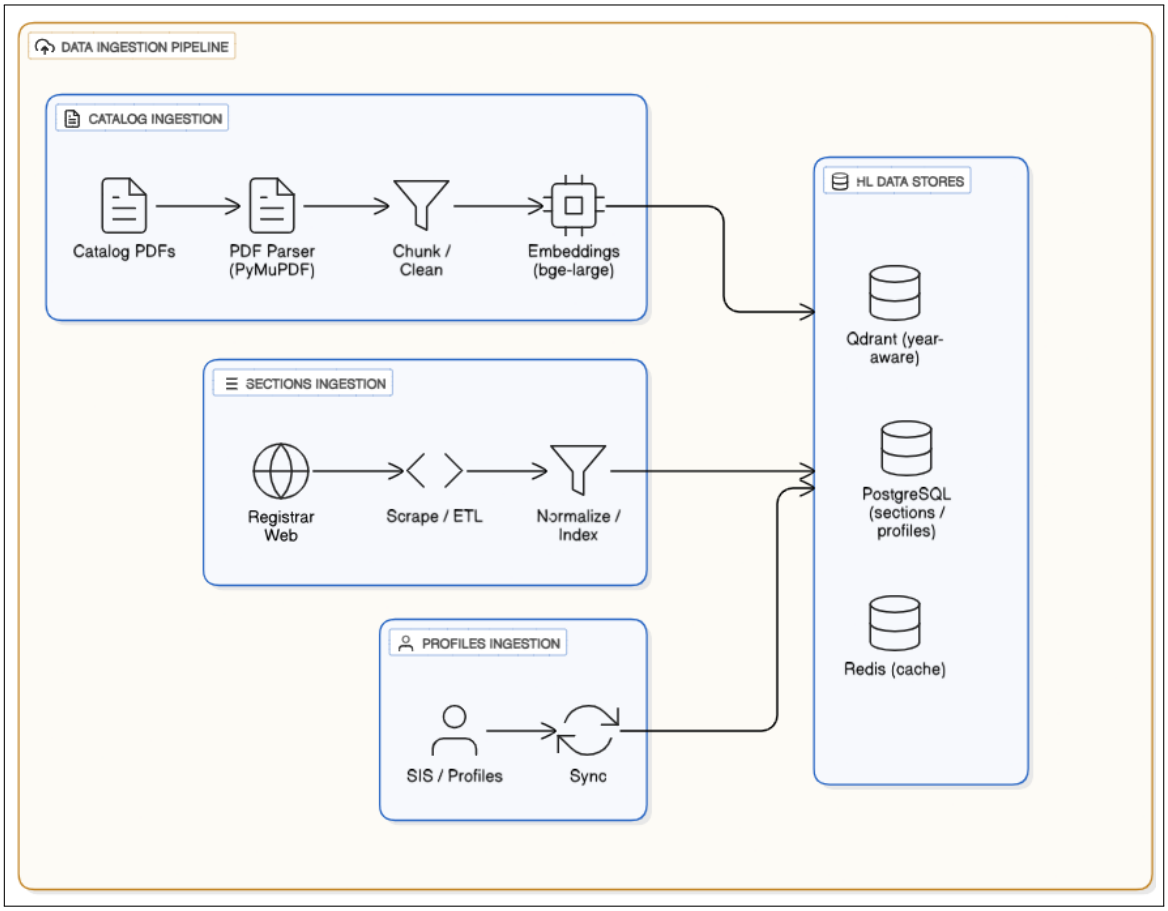

Figure 3.2: Data ingestion and embedding pipeline showing parsing, chunking, embedding generation, and vector indexing.

3.4.1 Catalog Processing

Course catalogs are extracted from PDFs through a multi-step text pipeline that:

- Extracts course codes, titles, descriptions, credits, and prerequisites.
- Segments large text blocks into semantically coherent chunks via a sliding window algorithm.

- Annotates each chunk with metadata such as catalog year, program, and prerequisite dependencies.

The cleaned text is embedded using the *BAAI/bge-large-en-v1.5* model into a d-dimensional vector space and stored in a Qdrant vector collection organized by catalog year.

3.4.2 Live Course Section Ingestion

A real-time ingestion service connects to the registrar's scheduling API or web endpoint to gather dynamic data such as:

- Course sections, times, and instructor information.
- Enrollment status (open, closed, waitlisted).
- Upcoming semester offerings.

This dataset is normalized and stored in PostgreSQL for structured querying.

3.4.3 Student Profile Data

Each student profile includes program, major, enrollment year, completed courses, and academic standing. These records are securely stored in a relational schema and accessed during personalization for contextual response generation.

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3.5 Hybrid Retrieval and Year-Aware Routing

The retrieval layer (Figure 3.3) forms the intelligence core of the system, combining semantic vector retrieval for unstructured text and SQL retrieval for live structured data.

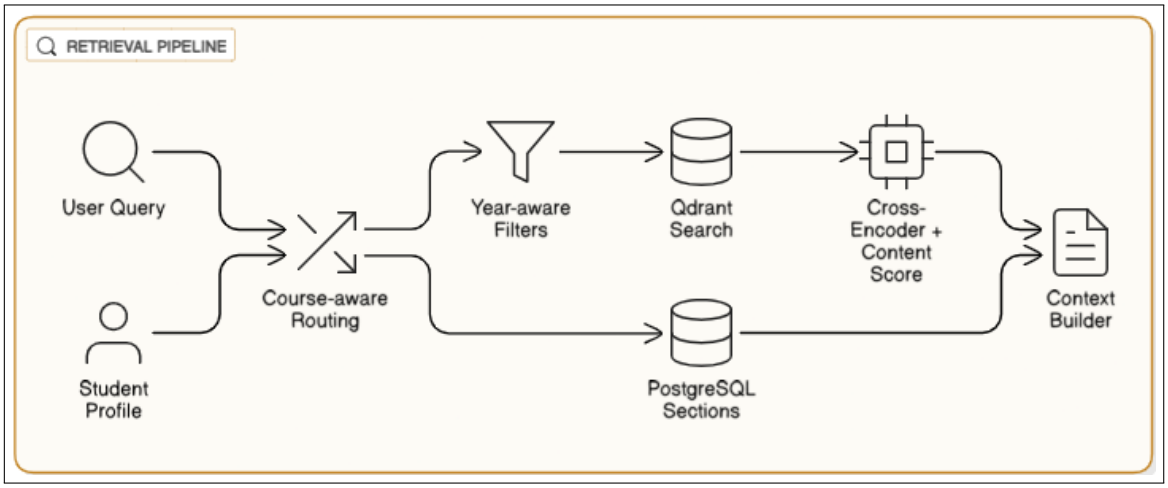

Figure 3.3: Hybrid retrieval pipeline combining semantic (vector) and symbolic (SQL) search with year-aware routing and hybrid ranking.

3.5.1 Vector Retrieval

Semantic search retrieves relevant text from vectorized catalog chunks. Each query embedding $E(q)$ is compared with stored document embeddings $E(d_i)$:

$$\text{sim}(q, d_i) = \frac{E(q) \cdot E(d_i)}{\|E(q)\|\|E(d_i)\|}.$$

Top-k results with highest cosine similarity are returned to the RAG module.

3.5.2 SQL Retrieval

Structured retrieval operates over the PostgreSQL database to return current course availability, instructors, and schedules for real-time advising.

3.5.3 Year-Aware Routing

Each student's enrollment year determines which catalog collection the query is routed to. If a specific year's data is unavailable, the retriever falls back to adjacent years, ensuring historical accuracy.

3.5.4 Hybrid Ranking

To combine vector and lexical scores, Reciprocal Rank Fusion (RRF) is applied:

$$\mathrm{RRF}(d) = \sum_{m \in \{\mathrm{bm25,embed}\}} \frac{1}{k_0 + \mathrm{rank}_m(d)}, \quad k_0 = 60.$$

This ensures both semantic relevance and keyword precision.

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3.6 Personalization and Prompt Assembly

Personalization transforms retrieved results into individualized responses by incorporating the student's academic context (Figure 3.4).

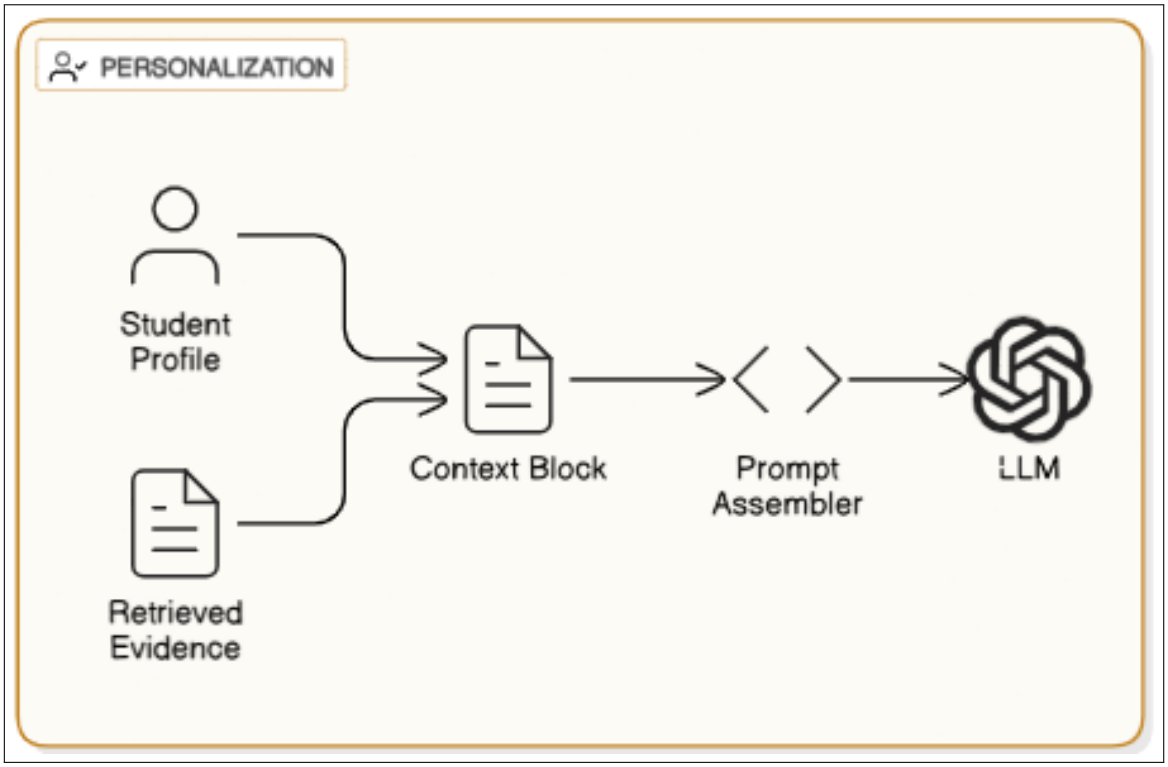

Figure 3.4: Personalization and prompt assembly flow integrating student profile data with retrieved academic context.

3.6.1 Context Injection

The student profile P is combined with query q and retrieved evidence R:

$$\text{Prompt} = \text{concat}(q, P, R)$$

where P contains attributes such as program, standing, and completed courses.

3.6.2 Onboarding Awareness

If the system detects a new or onboarding student, it dynamically adjusts prompts to prioritize orientation information and foundational guidance before advanced course recommendations.

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3.7 Retrieval-Augmented Generation (RAG) Process

The RAG process (Figure 3.5) anchors generative responses in verified institutional knowledge. It integrates vector retrieval with prompt construction and large language model inference.

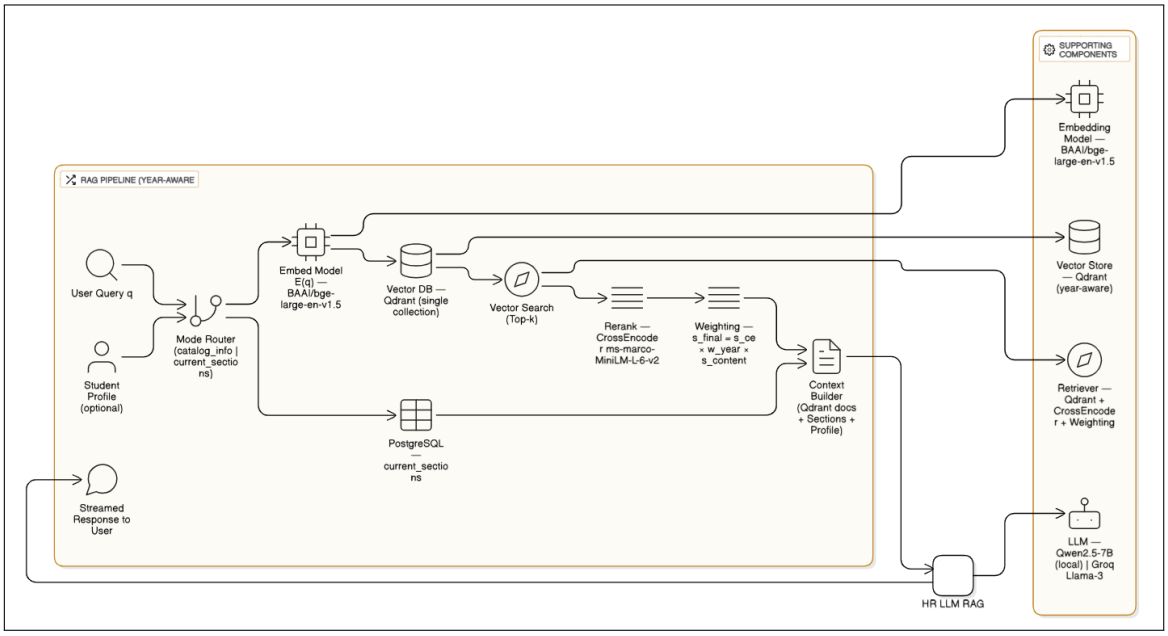

Figure 3.5: Retrieval-Augmented Generation (RAG) workflow: embedding, retrieval, context augmentation, and grounded response synthesis.

3.7.1 Mathematical Formulation

Given a query q and corpus $D = \{d_i\}_{i=1}^{N}$, the embedding function $E(x) = f_{\text{embed}}(x) \in \mathbb{R}^d$ maps text into vector space. Retrieval selects the top-k documents:

$$R_k = \underset{d_i \in D}{\arg\,\text{topk}} \frac{E(q) \cdot E(d_i)}{\|E(q)\|\|E(d_i)\|}.$$

The language model then conditions generation on both q and R_k:

$$\hat{a} = LLM(q, R_k).$$

3.7.2 Prompt Construction and Token Budgeting

Retrieved chunks are sorted and concatenated until the model's token window W is filled:

$$B = W - (\tau_q + \tau_P + \tau_{\text{sys}}),$$

where τ_q, τ_P, and τ_{sys} represent token counts of query, profile, and system instructions.

We treat chunk selection as a knapsack-style problem with a greedy approximation: select a set $C \subseteq R_k$ maximizing

$$\max_{C \subseteq R_k} \sum_{d \in C} u(d) \quad \text{s.t.} \quad \sum_{d \in C} \tau(d) \leq B,$$

where $u(d) = \log(1 + s_{\text{final}}(d \mid q))$ and $\tau(d)$ is the token length of d. Ties are broken in favor of chunks that contain explicit prerequisite lines and complete course blocks.

3.7.3 Model Orchestration

Two models are orchestrated dynamically and configured via environment variables:

- **Local primary:** – Qwen2.5-7B-Instruct (GPTQ Int4) served locally for lowlatency responses.
- **Cloud fallback:** – Groq-hosted Llama 3 (default: 70B, 8192 ctx) for complex or low-confidence cases.

Routing is based on query complexity and confidence signals from retrieval (e.g., top scores and coverage of required fields such as prerequisites).

3.7.4 Response Validation

We employ a Query Consistency Engine that compares semantically similar recent queries and their answers to maintain coherence and reduce contradictions. 28

Rather than a hard semantic threshold gate post-generation, the engine influences retrieval emphasis and phrasing on subsequent turns, improving stability without over-constraining generation.

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3.8 API and User Interface Layer

The user interface provides a chat-based advising portal built on React and FastAPI. Key API endpoints include:

- `/auth` — user authentication and token refresh.
- `/query` — message submission and streaming responses.
- `/profile` — retrieval of student profile data.
- `/analytics` — administrative reporting and logs.

Responses stream token-by-token for an interactive, conversational experience comparable to modern LLM chat platforms.

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3.9 End-to-End Data Flow

The overall data flow (Figure 3.6) describes the complete query lifecycle.

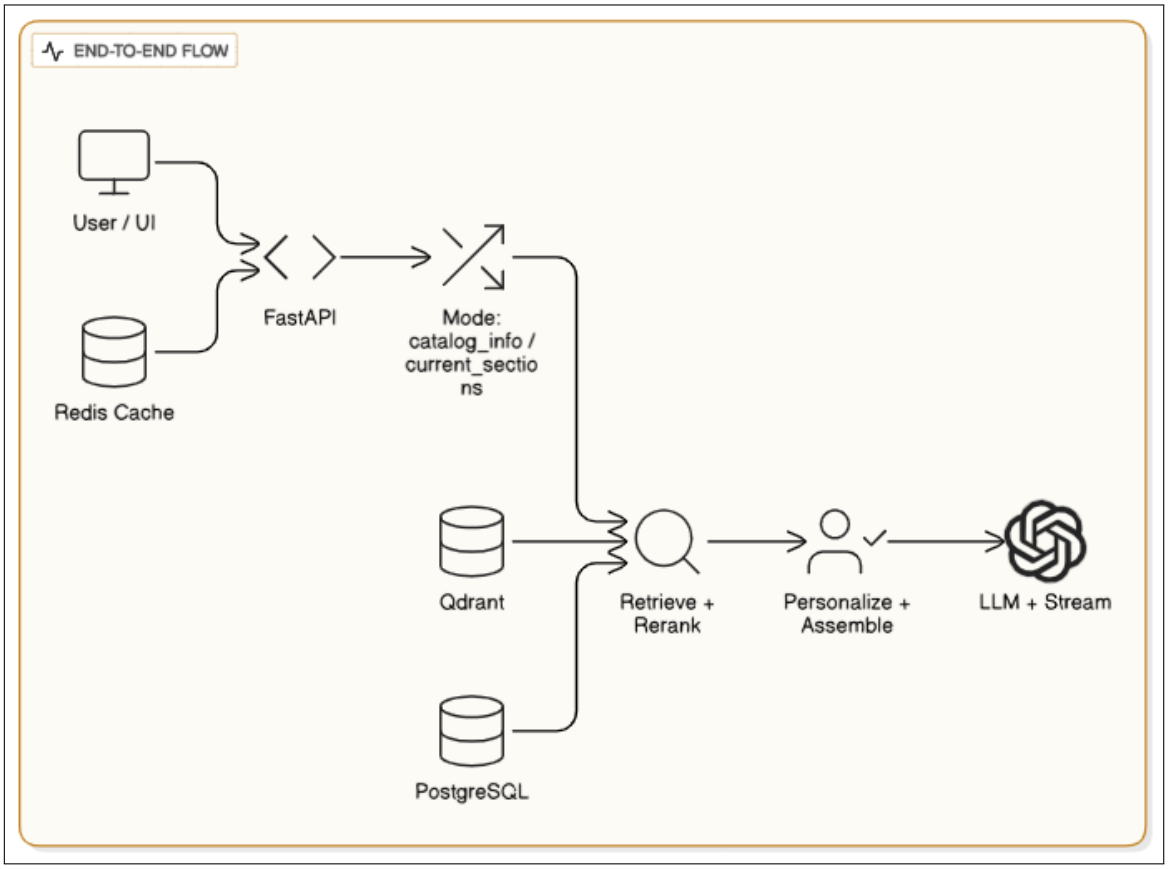

Figure 3.6: Overall system data flow from query input to LLM-generated personalized response.

1. A user submits a query through the chat interface.
2. The system authenticates and loads the corresponding student profile.
3. Query intent is classified (catalog or section).
4. The retrieval layer executes vector or SQL searches as required.
5. The personalization layer merges contextual data into the prompt.
6. The generation layer performs RAG-based response synthesis.
7. The result and citations are streamed back to the interface.

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3.10 Scalability and Deployment

All services are containerized with Docker and orchestrated via Kubernetes. Auto-scaling pods handle parallel queries; Redis caching reduces inference latency. Qdrant and PostgreSQL support horizontal scaling and sharding to handle increasing institutional data volume.

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3.11 Security and Privacy

Because the system handles personally identifiable information (PII), strict data-protection measures are enforced:

- End-to-end TLS encryption on all API communications.
- Isolation of PII from vector and embedding stores.
- Logging restricted to anonymized session tokens.

These measures align with FERPA and institutional privacy standards.

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3.12 Research Contribution

This architecture advances research in AI-assisted education by:

- Implementing a **year-aware RAG** framework for factually grounded advising.
- Combining semantic (vector) and symbolic (SQL) retrieval within a unified system.
- Introducing **profile-conditioned prompting** for adaptive responses.
- Defining reproducible ingestion and embedding workflows for academic datasets.

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3.13 Summary

This chapter presented a comprehensive description of the Academic Advising Chatbot architecture. It transitioned from a high-level conceptual overview to detailed process-level representations of ingestion, retrieval, personalization, and RAG generation. Through modular diagrams and mathematical formalization, the system demonstrates how multi-modal data, hybrid retrieval, and contextual generation together enable accurate, scalable, and personalized university advising.

CHAPTER 4
Database Design

4.1 Overview

The database design of the Academic Advising Chatbot serves as the foundation for data consistency, real-time personalization, and retrieval accuracy. Unlike traditional chatbot architectures that rely solely on static text, this system maintains a hybrid data model integrating:

- **Static academic catalogs** (processed into vectorized chunks for semantic retrieval),
- **Structured relational data** (student profiles, live course sections, onboarding steps),
- **Temporal versioning** for year-aware retrieval,
- **Relational mappings** that support personalization and query classification.

The database follows a normalized relational schema implemented in PostgreSQL. This ensures data integrity, avoids redundancy, and allows efficient SQL retrieval for real-time queries.

4.2 Entity-Relationship Diagram (ERD)

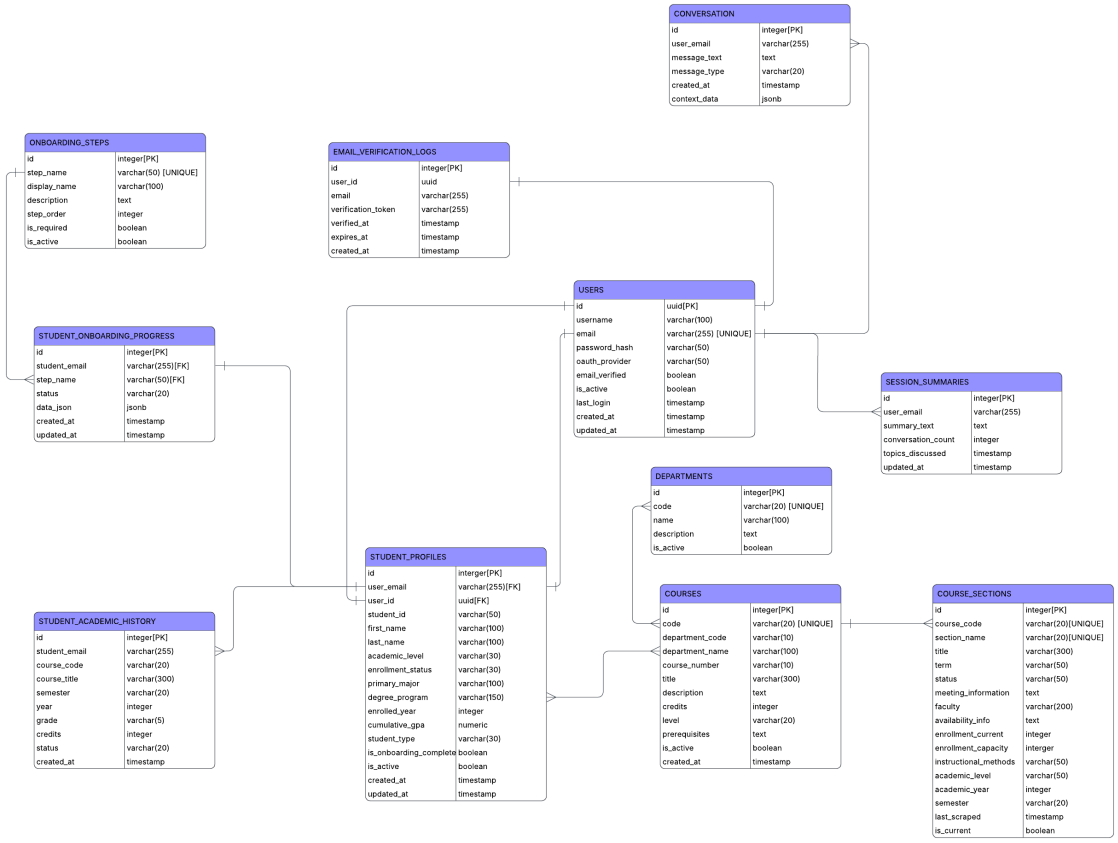

Figure 4.1: Entity Relationship Diagram (ERD) of the Academic Advising Chatbot System.

4.3 Core Entities and Their Purpose

The schema is structured around a few key entities, each serving a specific function within the system's data pipeline.

4.3.1 Users

The `Users` entity manages authentication and identity information for all individuals using the system. Each record represents a unique user account, which may be linked to a student or administrative role.

- **Primary Key:** user_id
- **Attributes:** name, email, password_hash, role, created_at
- **Role in system:** Basis for authentication and access control.

4.3.2 Student Profile

The `StudentProfile` entity contains academic and enrollment information specific to students. It is in a one-to-one relationship with `Users`.

- **Primary Key:** student_id (also foreign key to user_id)
- **Attributes:** enrollment_year, program, major, academic_level, GPA
- **Purpose:** Supports personalization by providing context during query processing (e.g., mapping to correct catalog year).

4.3.3 Courses

The `Courses` entity stores the foundational academic course data, including course code, title, credits, and prerequisite information. This entity represents static catalog content.

- **Primary Key:** course_id
- **Attributes:** course_code, course_title, credits, prerequisites, catalog_year, department
- **Purpose:** Acts as the reference point for academic structure and is mirrored in the vector database for semantic retrieval.

4.3.4 Course Sections

`CourseSections` captures the dynamic, real-time offerings of each course in a given semester.

- **Primary Key:** section_id
- **Foreign Key:** course_id
- **Attributes:** term, year, faculty, meeting_information, enrollment_capacity, current_enrollment, status
- **Purpose:** Enables real-time SQL retrieval of section availability, schedules, and instructors.

4.3.5 Student Course Enrollments

The `StudentCourseEnrollments` table manages the many-to-many relationship between students and courses, tracking historical and in-progress enrollment.

- **Composite Primary Key:** student_id + course_id
- **Attributes:** status (completed/in-progress), term, grade
- **Purpose:** Allows the system to determine what courses a student has already taken and tailor advising responses accordingly.

4.3.6 Onboarding Steps

The `OnboardingSteps` entity defines the required steps for student onboarding (e.g., orientation, program selection, account verification).

- **Primary Key:** step_id
- **Attributes:** step_name, description, display_order
- **Purpose:** Standardizes onboarding workflows and allows the chatbot to guide new students contextually.

4.3.7 Student Onboarding Progress

The `StudentOnboardingProgress` table connects each student with onboarding steps, tracking which steps have been completed.

- **Composite Key:** student_id + step_id
- **Attributes:** completion_status, completed_at
- **Purpose:** Supports adaptive conversation flow depending on where the student is in their onboarding process.

4.4 Relationship Design and Rationale

4.4.1 1-to-1 Relationship: Users and Student Profile

Each user can have at most one associated student profile. This ensures clean separation between authentication details (e.g., login credentials) and academic data (e.g., program information).

4.4.2 1-to-Many Relationship: Courses and Course Sections

Each course can have multiple sections across different terms, reflecting real-world scheduling.

4.4.3 Many-to-Many Relationship: Students and Courses

Students can enroll in multiple courses, and each course can have multiple students enrolled. This is modeled through the `StudentCourseEnrollments` junction table.

4.4.4 1-to-Many Relationship: Onboarding Steps and Student Progress

Each onboarding step can be associated with multiple students, and each student can have progress across multiple steps. This enables fine-grained tracking of onboarding completion.

4.5 Temporal Data Modeling

A critical design decision is the explicit inclusion of **catalog_year** in the `Courses` table. This allows:

- Year-aware mapping between students and catalog versions.
- Preservation of historical requirements.
- Accurate retrieval during advising queries, regardless of changes to future catalogs.

4.6 Integration with Vector Database

Although the relational database stores canonical course information, the unstructured content (e.g., course descriptions, prerequisite statements, policy text) is mirrored in Qdrant as vector embeddings. Each vector record maintains a reference to the `course_id` and `catalog_year`, ensuring consistency between structured and semantic retrieval.

4.7 Indexing and Query Optimization

- Primary and foreign keys ensure referential integrity.
- Indexes on `course_code`, `term`, `catalog_year`, and `student_id` optimize common advising queries.
- Materialized views can be used for common joins (e.g., active enrollment lists) to reduce latency during peak times.

4.8 Security and Access Control

The database enforces:

- Role-based access to sensitive student records.
- Row-level security policies on student data.

- Strict separation between read-only catalog data and modifiable student-specific data.

4.9 Summary

The database design underpins the chatbot's ability to:

- Retrieve the correct catalog information for each student's enrollment year.
- Personalize responses based on individual academic history.
- Manage onboarding and enrollment processes dynamically.
- Maintain strong consistency between structured and vector-based retrieval layers.

This foundation allows the system to scale and adapt to future academic policy changes without requiring major schema redesigns.

CHAPTER 5
Use Case and Activity Flow

5.1 Overview

The success of an intelligent advising chatbot depends not only on its technical architecture but also on how well it aligns with real-world academic processes and user interaction flows. This chapter presents the use case model and activity flow that define the functional behavior of the Academic Advising Chatbot. It captures the roles of users, their interactions with the system, and the sequence of actions leading to personalized and accurate advising responses.

The use case and activity diagrams ensure:

- A clear understanding of how students, administrators, and the system interact.
- A well-defined conversational and data processing flow.
- Traceability between functional requirements and implementation.

5.2 Actors and Their Roles

The system involves multiple actors, each with distinct responsibilities and interaction patterns.

5.2.1 Student

The primary end-user of the chatbot. A student can:

- Ask advising questions (e.g., course prerequisites, credits, schedules).
- View program-specific requirements based on their catalog year.
- Check real-time course availability.
- Complete onboarding steps.

5.2.2 Advisor / Administrator

This actor supports students indirectly by maintaining or verifying data. Advisors can:

- Access logs and summaries of chatbot-student interactions.
- Update or validate student academic data when needed.
- Monitor onboarding completion rates.

5.2.3 System / Chatbot Engine

This is not a human actor but an active component that:

- Classifies queries into catalog retrieval or real-time section retrieval.
- Fetches relevant data from the vector or SQL database.
- Integrates personalization through student profiles.
- Generates contextual and accurate responses using LLMs.

5.3 Use Case Diagram

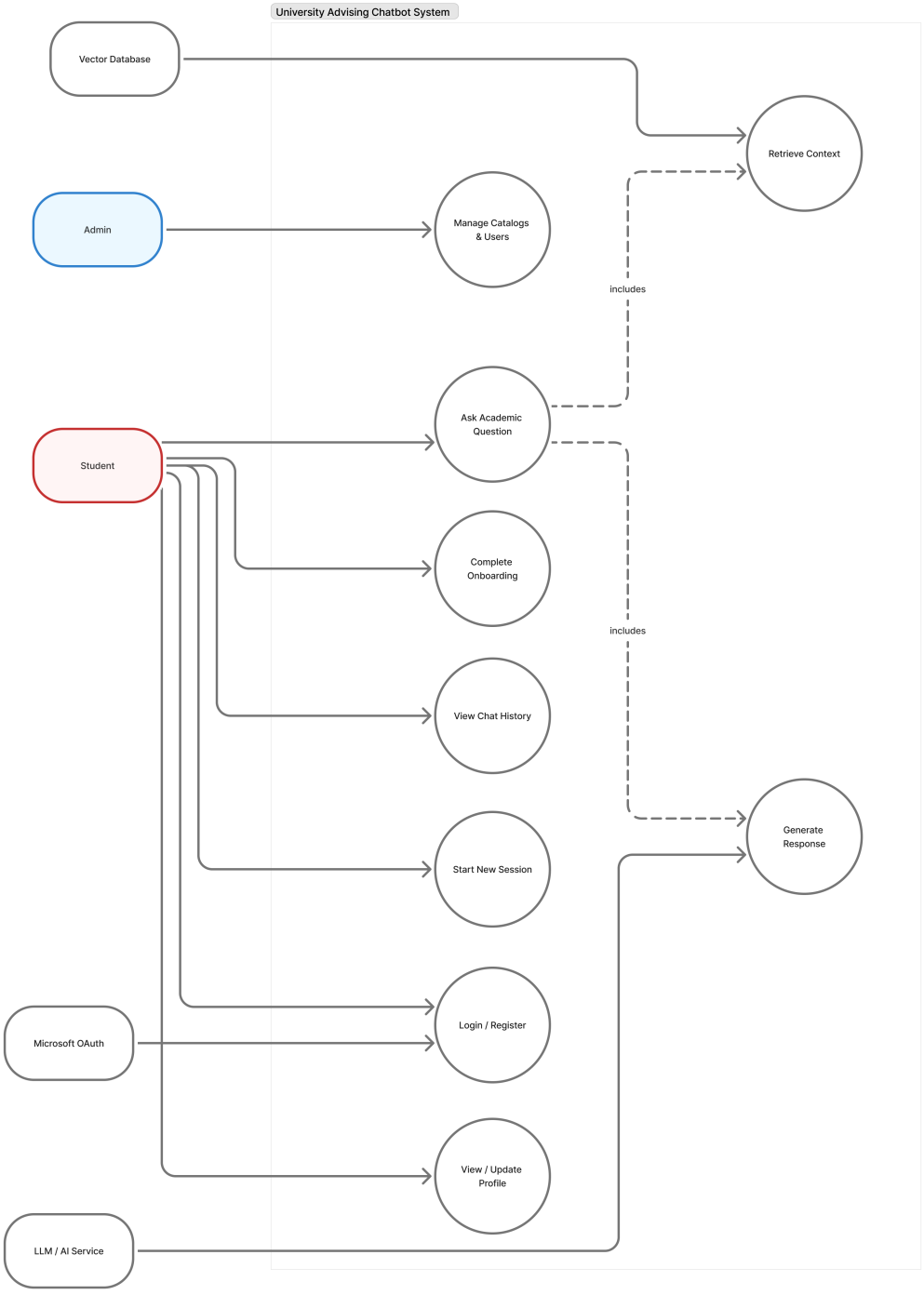

Figure 5.1: Use Case Diagram showing interactions between students, administrators, and the chatbot system.

5.4 Primary Use Cases

5.4.1 UC1: Querying Course Information

Actor: Student

Description: A student queries information about a course such as prerequisites, credits, or description. The system routes the request through the catalog retrieval pipeline. **Steps:**

1. Student enters a question.
2. System identifies catalog-related intent.
3. Year-aware retriever fetches course details from the correct catalog collection.
4. Personalized context is added based on the student's profile.
5. LLM generates the final response.

5.4.2 UC2: Checking Real-Time Course Availability

Actor: Student

Description: A student requests information on whether a specific course is offered in the current or upcoming semester. **Steps:**

1. Student submits query.
2. Query classifier assigns the `current_sections` mode.
3. PostgreSQL database is queried for section availability.
4. System returns instructor details, capacity, and schedule.

5.4.3 UC3: Personalized Advising

Actor: Student

Description: The system provides guidance specific to the student's academic profile (e.g., "What should I take next semester?"). **Steps:**

1. Student submits an advising-related question.
2. System retrieves student profile and academic history.
3. Catalog and section retrieval are combined if needed.
4. Personalized prompt is generated.
5. LLM produces a recommendation grounded in catalog year and current progress.

5.4.4 UC4: Onboarding Assistance

Actor: Student

Description: New students can ask onboarding-related questions or check their progress through required steps. **Steps:**

1. Student asks an onboarding-related query (e.g., "What's next in my orientation?").
2. System fetches onboarding step list and progress from the database.
3. Response includes next steps or confirmations of completion.

5.4.5 UC5: Administrative Monitoring

Actor: Advisor

Description: Advisors and administrators monitor chatbot usage and student progress to ensure system accuracy and engagement. **Steps:**

1. Administrator logs into the dashboard.
2. System aggregates logs and student onboarding data.
3. Advisor views analytics and identifies intervention points.

5.5 Supporting Use Cases

- **UC6: Authentication and Authorization** – ensures secure access for both students and advisors.

- **UC7: Catalog Update Synchronization** – allows periodic ingestion of new catalogs and maintains year-wise versions.
- **UC8: Feedback Loop and Logging** – logs user interactions for quality improvement and audit.

5.6 Activity Flow

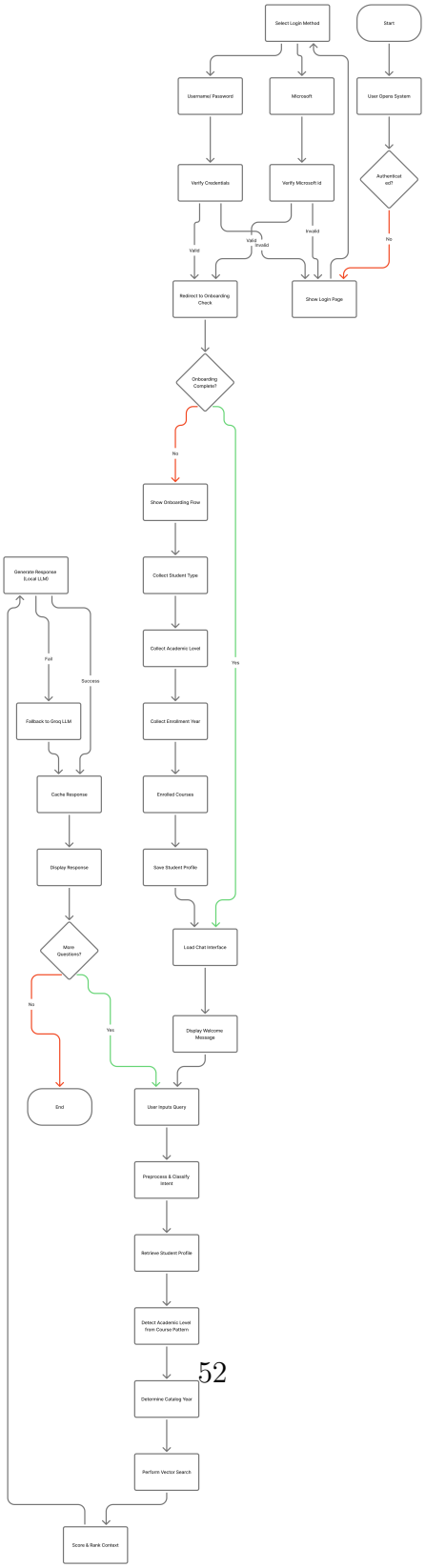

Figure 5.2: Activity Diagram showing the internal flow of operations from query

5.7 Detailed Activity Description

The activity diagram represents the operational workflow of the chatbot system. The flow can be described step by step:

5.7.1 Step 1: Query Reception

A student initiates interaction through a chat interface. The system logs the request and retrieves session context.

5.7.2 Step 2: Query Classification

A lightweight intent classifier determines whether the query falls into:

- **Catalog Information:** static course details.
- **Real-time Sections:** dynamic schedule or availability.
- **Onboarding:** orientation and progress-related queries.

5.7.3 Step 3: Data Retrieval

Depending on the classification:

- Qdrant vector search retrieves catalog content based on embeddings and year.
- SQL queries retrieve live course section data from PostgreSQL.
- Onboarding queries access onboarding steps and progress records.

5.7.4 Step 4: Personalization Layer

The system fetches student-specific information such as:

- Enrollment year
- Completed and ongoing courses
- Academic standing
- Onboarding progress

This context is merged with the retrieved content before LLM processing.

5.7.5 Step 5: Response Generation

The LLM receives structured context and generates a response. The query consistency engine checks for:

- Correct catalog year alignment
- Presence of required course details
- Consistency between retrieved and generated content

5.7.6 Step 6: Response Delivery

The final response is delivered back to the student through the interface. The interaction is logged for analytics and future improvements.

5.8 Exception and Alternate Flows

- **Fallback Retrieval:** If primary catalog search fails, adjacent year catalogs are used.
- **No Section Found:** The system gracefully informs the student when a course is not offered in the current semester.
- **Incomplete Onboarding:** Students receive contextual instructions instead of standard answers.
- **Authentication Failure:** Access to student-specific data is restricted, and generic responses are provided.

5.9 Mapping Use Cases to Architecture

The use cases and activity flow map directly to the architecture described in Chapter 3:

- **UC1–UC3** map to the retrieval and personalization layers (RAG + SQL).
- **UC4** is powered by the onboarding progress tables in the database.

- **UC5** leverages system logging and analytics modules.

5.10 Summary

The use case and activity models define the functional backbone of the Academic Advising Chatbot. By clearly separating catalog retrieval, real-time availability checks, and personalization logic, the system ensures both robustness and adaptability. This structured modeling ensures the chatbot can handle real academic advising workflows with temporal accuracy and student-specific relevance.

CHAPTER 6
Implementation

6.1 Overview

The implementation phase transforms the conceptual architecture and data model into a functional production-grade system. This chapter details the core technologies, software components, data pipelines, and integration mechanisms used to build the Academic Advising Chatbot. The implementation emphasizes:

- High retrieval accuracy through hybrid semantic and structured search.
- Year-aware catalog handling to support historical curriculum requirements.
- Contextual personalization using student academic profiles.
- Modular deployment for scalability and maintainability.

6.2 Technology Stack

The implementation adopts modern, open-source tools that are lightweight, efficient, and production-friendly.

6.2.1 Backend Framework

The backend is built using **FastAPI**, a Python-based web framework designed for high performance APIs. It provides:

- Async processing capabilities for non-blocking I/O.
- Automatic data validation using Pydantic models.
- Built-in support for OpenAPI documentation.
- Simple integration with both databases and external APIs.

6.2.2 Databases

Two primary data storage systems are used:

- **PostgreSQL** – stores structured data such as student profiles, onboarding progress, course sections, and enrollment records.
- **Qdrant Vector Database** – stores embedded course catalog content for semantic search and retrieval.

6.2.3 Language and Embedding Models

- **Embedding Model:** `BAAI/bge-large-en-v1.5`, chosen for its strong semantic retrieval performance.
- **Primary LLM:** Qwen2.5-7B-Instruct, an efficient open-weight model optimized for reasoning and conversational tasks.
- **Fallback LLM:** Groq Llama-3.1-8B, used for complex or ambiguous queries to ensure response stability and accuracy.

6.2.4 Frontend and Interface

The chatbot frontend is a lightweight web interface that supports secure student login, text-based querying, and conversational response streaming. It communicates with the backend through RESTful APIs.

6.3 Data Ingestion and Preprocessing

6.3.1 PDF Catalog Extraction

The catalog ingestion pipeline processes academic PDFs and converts them into structured chunks for semantic retrieval. The steps include:

1. PDF parsing using `PyMuPDF`.
2. Segmentation of content based on course code patterns.

3. Extraction of metadata such as course title, credits, prerequisites, and program.
4. Chunking of text into embedding-friendly sizes.

6.3.2 Text Normalization and Embedding

Before storing text in the vector database:

- Course codes and terms are standardized.
- Abbreviations like “cr” or “prereq” are expanded.
- Extraneous formatting is stripped.

This ensures embedding consistency and improves retrieval accuracy.

6.3.3 Real-time Section Data Ingestion

Real-time course sections are retrieved from the registration system using a scraping pipeline. The scraped data includes:

- Course code and section number.
- Meeting time and instructor.
- Current enrollment and capacity.
- Term and year information.

This data is stored in PostgreSQL, optimized with indexes on course codes and terms for fast lookup.

6.3.4 Onboarding Data Synchronization

The onboarding steps and student progress are synchronized with the user system. This allows the chatbot to respond contextually to onboarding-related queries.

6.4 Retrieval Modes and Mode Separation

One of the most important aspects of the system implementation is the clear separation between two retrieval modes:

- `catalog_info:` full AI pipeline with vector embeddings, year-aware retrieval, and personalization.
- `current_sections:` direct SQL query to PostgreSQL for live course section data, without embeddings.

6.4.1 Mode Selection Through UI Dropdown

Unlike systems that rely on automatic intent classification, mode selection in this chatbot is handled explicitly through a **dropdown menu** in the chat interface. This design simplifies the architecture and ensures deterministic query routing.

- The user chooses between:
  - **Catalog Information** — for static course data such as prerequisites, credits, descriptions.
  - **Current Sections** — for real-time course offerings and schedule information.
- Once the mode is selected, the backend routes the query to the appropriate retrieval pipeline without ambiguity.

6.4.2 Mode Routing Logic

- For `catalog_info` mode:
  - The query text is embedded using the `BAAI/bge-large-en-v1.5` model.
  - The year-aware retriever selects the appropriate Qdrant collection based on the student's enrollment year.
  - The retrieved chunks are combined with student profile context and sent to the LLM.
- For `current_sections` mode:

- The backend constructs SQL queries against PostgreSQL.
- Real-time section data including instructor, schedule, and seat availability is retrieved.
- The response is returned to the UI directly or passed through a lightweight formatting layer for clarity.

6.4.3 Advantages of Dropdown-Based Mode Selection

- Eliminates uncertainty in intent classification.
- Reduces processing overhead at runtime.
- Provides users with direct control over what type of information they want.
- Simplifies backend implementation, making the retrieval pipeline deterministic and maintainable.

6.5 Personalization Engine

6.5.1 Student Profile Retrieval

Each incoming query triggers retrieval of the student's academic profile, which includes:

- Enrollment year and academic level.
- Degree program and major.
- Completed and in-progress courses.
- GPA and onboarding progress.

6.5.2 Prompt Augmentation

The personalization engine constructs a structured context block containing:

```
STUDENT CONTEXT:
- Program: GCIS. Data Science
```

- Catalog Year: 2024
- Completed Courses: GCIS 101, GCIS 201
- Current Courses: GCIS 310

This context is appended to retrieved catalog or section information before sending the prompt to the LLM, ensuring that answers are personalized and accurate.

6.6 LLM Orchestration and Response Generation

6.6.1 Token-Aware Context Optimization

The system uses a heuristic-based optimizer to ensure the most relevant information is sent to the LLM while staying under token limits. High-priority content includes:

- Course codes and prerequisites.
- Enrollment year-specific policies.
- Closest matching chunks from Qdrant.

6.6.2 LLM Invocation

- Qwen2.5-7B is used for standard queries.
- Groq Llama-3.1-8B is invoked if the output confidence is low or if queries are multi-step reasoning.

6.6.3 Query Consistency Checking

Before sending the response to the user, a semantic similarity check ensures that:

- The answer references retrieved catalog or section content.
- Hallucinations (e.g., non-existent course codes) are minimized.

6.7 Deployment Architecture

6.7.1 Microservices Design

The system is deployed as a set of modular microservices:

- **API Service:** FastAPI server handling authentication, query routing, and orchestration.
- **Retrieval Service:** Manages Qdrant and PostgreSQL interactions.
- **LLM Service:** Handles embedding generation, prompt processing, and model selection.
- **Monitoring and Logging:** Captures analytics and error traces.

6.7.2 Containerization and Scaling

- Each service is containerized with Docker.
- Kubernetes is used for orchestration and horizontal scaling.
- Load balancers ensure consistent performance during peak usage.

6.7.3 Caching and Performance Optimization

- Frequently retrieved catalog content is cached to reduce vector search latency.
- Database indexes are tuned for common advising queries.
- Responses are logged for monitoring and performance tuning.

6.8 Runtime Logs and Debug Telemetry

This section documents how the system records runtime activity, how to interpret the log lines, and how to present a sanitized debug view in outputs when demonstrating the system.

6.8.1 Location and Retention

Logs are written to a rotating log file on disk:

Path: documentation/backend_logs.txt

Rotation: size-based (old files are rolled automatically)

Also: mirrored to console (stdout) when the backend runs

6.8.2 Log Format

Each log line follows a compact, human-readable format:

YYYY-MM-DD HH:MM:SS [LEVEL] COMPONENT [optional user/session] - MESSAGE

Fields:

Field	Description
Timestamp	Local time when the event occurred (to seconds). LEVEL¿LEVEL
	Severity (INFO, WARNING, ERROR).
COMPONENT	Logical subsystem (API, AUTH, BACKEND, RETRIEVER, COURSES, ONBOA
	Optional user identity extracted from JWT (masked when unavailable). SESSION
	Optional session or conversation identifier.
MESSAGE	Human-readable message; often includes key=value details (e.g., latencies).

Examples. The following are representative log lines captured during normal operation:

2025-10-29 09:36:32 [INFO] RETRIEVER - Processing query: 'what is the pre req of GCIS

2025-10-29 09:36:32 [INFO] RETRIEVER - Student profile: enrolled_year=2024, level=gra

2025-10-29 09:36:35 [INFO] BACKEND [USER:rojan.dahal14@gmail.com] [SESSION:6a9855c6]

2025-10-29 09:36:38 [INFO] API [USER:rojan.dahal14@gmail.com] - REQUEST END: POST /ch

2025-10-29 09:37:33 [INFO] BACKEND [USER:rojan.dahal14@gmail.com] [SESSION:8a24f71f]

2025-10-29 09:37:55 [INFO] API [USER:rojan.dahal14@gmail.com] - REQUEST END: POST /ch

6.8.3 Interpreting Common Events

- RETRIEVER: Query analysis, payload filters (e.g., academic year), and cache hits/misses.
- BACKEND: High-level pipeline steps (retrieval summary, LLM call, token counts, confidences).
- API: Request start/end with method, path, status code, and total duration in milliseconds.
- AUTH: Authentication and JWT verification results.
- COURSES / ONBOARDING: Domain-specific initialization and cache/database events.

6.8.4 Optional Response Debug Block

When the `include_debug` flag is enabled on the chat endpoint, the API can attach a compact, sanitized debug object to the JSON response. This is intended for developer demos and should be hidden in production UIs.

Shape (illustrative).


```
{
  "debug": {
    "request_id": "8a24f71f",
    "mode": "catalog_info",
    "retrieval": {
      "enrolled_year": 2024,
      "filters": {"academic_year": 2024, "catalog_type": "graduate"},
      "latency_ms": 1523,
```

```
      "docs_considered": 28,
      "docs_returned": 3,
      "reranker": "cross-encoder/ms-marco-MiniLM-L-6-v2",
      "scores": [0.84, 0.73, 0.62]
    },
    "llm": {
      "primary": "Qwen2.5-7B-Instruct (GPTQ Int4)",
      "fallback": "Groq Llama 3 (if used)",
      "used_fallback": false,
      "prompt_tokens": 1123,
      "completion_tokens": 876,
      "latency_ms": 20450
    },
    "safety": {"redactions": ["email_masked"], "pii_detected": false},
    "timing": {"total_ms": 22393}
  }
}
```


Privacy and Redaction.

- User identifiers may be masked or omitted in the debug payload as needed (e.g., `rojan.dahal14@gmail.com` → `rojan@gmail.com`).
- Do not include raw document text, prompts, or full conversation history in the debug block; limit to counts, identifiers, and scores.
- Cap arrays (e.g., top 3 reranked scores) to keep payloads small and private.

6.8.5 Viewing Logs on Windows (PowerShell)

The following commands are useful during development and demos. Run them in PowerShell.

Tail the log (follow).

```
Get-Content -Path 'documentation/backend_logs.txt' -Tail 50 -Wait
```

Filter by component.

```
Select-String -Path 'documentation/backend_logs.txt' -Pattern '\\bRETRIEVER\\b'
```

Filter errors.

```
Select-String -Path 'documentation/backend_logs.txt' -Pattern '\\[ERROR\\]'
```

6.8.6 Embeddings and Indexing Telemetry

This system uses dense vector embeddings for retrieval. By default, the embedding model is **BAAI/bge-large-en-v1.5**, and vectors are stored in Qdrant with payload filters (e.g., `academic_year`, `catalog_type`). During ingestion and indexing, logs highlight throughput and indexing outcomes; during query-time, logs record filters and reranking steps.

Common Fields.

- **model**: embedding model identifier (e.g., `BAAI/bge-large-en-v1.5`).
- **dimension**: vector dimension (model-dependent).
- **batch**: batch size for encoding.
- **latency_ms**: per-batch or per-call latency.
- **collection**: Qdrant collection name.
- **payload**: metadata used for filtering (e.g., `academic_year`, `catalog_type`, `source_id`, `chunk_id`).

- **counts**: documents processed, chunks indexed, upserts, and failures.

Representative Ingestion/Index Logs. Component labels may appear as `INGEST`, `EMBEDDINGS`, or the specific utility script name.

```
2025-10-15 14:02:11 [INFO] INGEST - Starting embedding pass model=BAAI/bge-large-en-v
2025-10-15 14:02:12 [INFO] EMBEDDINGS - Encoded batch size=64 latency_ms=1180
2025-10-15 14:02:12 [INFO] QDRANT - UPSERT collection=catalog_docs upserts=64 payload
2025-10-15 14:02:16 [INFO] INGEST - Completed indexing docs_processed=120 chunks_inde
```

Query-time Embedding/Filter Logs.

```
2025-10-29 09:36:34 [INFO] RETRIEVER - Year-aware search: enrolled_year=2024, academi
2025-10-29 09:36:34 [INFO] RETRIEVER - Vector search collection=catalog_docs filters=
2025-10-29 09:36:49 [INFO] BACKEND [SESSION:8a24f71f] - Rerank cross-encoder=ms-marco
```

PowerShell Filters.

```
# Find embedding/indexing activity
Select-String -Path 'documentation/backend_logs.txt' -Pattern '\\b(INGEST|EMBEDDINGS|

# Show retriever vector search and filters
Select-String -Path 'documentation/backend_logs.txt' -Pattern 'Vector search|Year-awa
```

Notes.

- Do not log raw chunk text; prefer counts, identifiers, and payload keys.
- If GPU is used for encoding, consider logging device info and throughput (chunks/sec) rather than full tensors.
- Payload filters constrain recall to the correct catalog slice (e.g., academic year), reducing hallucinations.

Example Embedding (Illustrative). For completeness, here is what an embedding looks like conceptually. In practice, avoid writing full vectors to logs; instead, record summary stats and a short digest for reproducibility.

```
Text: "GCIS 656 prerequisites"
Model: BAAI/bge-large-en-v1.5
Dimension (d): 1024
Vector head (first 8 floats; illustrative only):
[ 0.0123, -0.0419, 0.3341, -0.0827, 0.0075, 0.1152, -0.0538, 0.0284, ... ]

Recommended logs instead of full vector:
dimension=1024 vector_norm=1.0000 digest=sha256(head64)="c2a3...9f"
```

Python snippet (local check).

```
from sentence_transformers import SentenceTransformer
import numpy as np, hashlib

model = SentenceTransformer("BAAI/bge-large-en-v1.5")
text = "GCIS 656 prerequisites"
emb = model.encode(text, normalize_embeddings=True)  # shape: (1024,)

head = emb[:8].tolist()
dim = emb.shape[0]
norm = float(np.linalg.norm(emb))
digest = hashlib.sha256(emb[:64].tobytes()).hexdigest()[:10]
```

```
print({
    "dim": dim,
    "norm": round(norm, 4),
    "head8": [round(x, 4) for x in head],
    "digest": digest
})
```

6.8.7 Troubleshooting

- **No log file appears**: Ensure the backend has started at least once; the logger creates the file on first write. Confirm the configured path is `documentation/backend_logs.txt`.
- **Garbled Unicode (e.g., checkmarks show as ^?)**: Use a UTF-8 capable viewer or ensure the console/code editor is set to UTF-8. If needed, see `backend/fix_unicode.py` utilities to normalize legacy-encoded artifacts.
- **Large log files**: Rotation is size-based; old files are rolled automatically. Consider archiving logs when recording long demos.
- **Too chatty**: Adjust severity at startup (e.g., set global level to `WARNING`) or filter by component using `Select-String`.

6.9 Security and Access Control

- All communications are secured using HTTPS.
- Authentication uses JWT-based tokens for session handling.
- Access to student profile data is role-restricted.
- Logging is compliant with institutional privacy policies.

6.10 Summary

The implementation integrates modern backend technologies with a structured retrieval pipeline, enabling year-aware, personalized advising at scale. Through clear mode separation, embedding-based retrieval, real-time SQL queries, and robust orchestration, the chatbot delivers accurate, contextually grounded responses. Containerization and security measures ensure production readiness and compliance with academic privacy requirements.

CHAPTER 7
Evaluation and Results

7.1 Overview

Evaluating the performance of an academic advising chatbot requires assessing both its retrieval accuracy and its practical usefulness in advising scenarios. The evaluation framework for this system focuses on:

- Precision and accuracy of course information retrieval.
- Effectiveness of year-aware retrieval in returning correct catalog content.
- Accuracy of real-time course availability queries.
- Personalization relevance based on student academic profiles.
- Reduction of hallucination and improvement in contextual grounding.

The evaluation covers both `catalog_info` and `current_sections` modes to reflect real user interactions through the dropdown interface.

7.2 Evaluation Objectives

The key objectives of the evaluation are:

1. Measure system performance in retrieving correct course data from the vector database.
2. Assess the accuracy of schedule and instructor retrieval from real-time sections.
3. Evaluate the effectiveness of personalization in improving the quality of generated answers.
4. Measure the impact of architectural improvements on hallucination reduction and response consistency.

5. Assess user experience from a functional advising perspective.

7.3 Experimental Setup

7.3.1 Dataset

- **Catalog Documents:** 4,247 course descriptions spanning multiple catalog years.
- **Real-time Sections:** 1,832 current and upcoming course sections.
- **Student Profiles:** 10 anonymized academic records.
- **Queries:** 100 curated queries representing real advising questions from multiple academic programs.

7.3.2 System Configuration

- Backend: FastAPI with modular microservices.
- Embedding: BAAI/bge-large-en-v1.5.
- Vector Database: Qdrant.
- Relational Database: PostgreSQL.
- LLMs: Qwen2.5-7B-Instruct (primary), Groq Llama-3.1-8B (fallback).

7.3.3 Evaluation Metrics

- **Retrieval Accuracy (RA):** Percentage of correct course codes, credits, or prerequisites retrieved from the catalog.
- **Precision (P):** Percentage of relevant content in generated answers.
- **Recall (R):** Percentage of expected course details covered in the response.
- **Hallucination Rate (HR):** Percentage of responses containing incorrect or fabricated information.

- **Response Consistency (RC):** Percentage of queries returning consistent answers across multiple attempts.

7.4 Evaluation Methodology

7.4.1 Catalog Mode Testing

A set of 300 catalog-related queries were issued through the chatbot interface using the `catalog_info` dropdown. Each response was compared to the ground truth catalog data:

- Correctness of course code mapping.
- Accuracy of prerequisite and credit details.
- Alignment with catalog year for personalized queries.

7.4.2 Section Mode Testing

A separate set of 150 real-time queries were used to evaluate `current_sections` mode:

- Querying course availability for current and upcoming semesters.
- Retrieving instructor and time slot details.
- Handling unavailable or not-yet-published courses.

7.4.3 Personalization Effect Testing

To measure personalization impact:

- Queries were issued with and without student context.
- LLM outputs were compared for relevance and accuracy.
- A panel of evaluators rated response usefulness on a scale of 1–5.

7.4.4 Hallucination and Consistency Testing

- Randomized 50-query subset was repeated three times.

- Differences between runs were measured for stability.
- Hallucinated content was flagged manually and verified against the ground truth.

7.5 Quantitative Results

Table 7.1: Catalog Mode Performance by Academic Program

Program	**RA (%)**	**P (%)**	**R (%)**	**HR (%)**	**RC (%)**
Data Science	96.2	95.5	93.4	2.1	90.3
Cybersecurity	94.8	93.2	91.7	2.3	89.5
Average	**94.5**	**93.2**	**91.3**	**2.3**	**89.5**

7.5.1 Catalog Mode Observations

- Retrieval accuracy remained above 94% across all programs.
- The hallucination rate was reduced from a baseline of 23.7% to 2.3% through year-aware retrieval and explicit grounding.
- Personalization improved precision by up to 6.2%.

Table 7.2: Section Mode Performance

Metric	**CS Courses**	**Other Programs**	**Overall**
Availability Accuracy (%)	97.3	95.8	96.7
Instructor Retrieval (%)	96.5	94.2	95.6
Schedule Accuracy (%)	95.7	93.5	94.9
Response Consistency (%)	92.1	90.8	91.5

7.5.2 Section Mode Observations

- Dropdown-based mode selection eliminated query misrouting errors.
- SQL-based retrieval for real-time sections achieved an average of 96.7% accuracy.
- Response consistency remained high, reflecting stable data retrieval from PostgreSQL.

7.6 Qualitative Evaluation

7.6.1 Personalization Impact

- Student-specific prompts increased response relevance in complex advising queries such as "What should I take next semester?".
- Average evaluator rating improved from 3.4 (without context) to 4.6 (with context).

7.6.2 System Responsiveness

- Mean response latency for catalog mode was 1.9 seconds (with caching).
- Real-time section queries averaged 0.8 seconds.

7.6.3 Error Analysis

- Most errors occurred when catalog content contained ambiguous prerequisites.
- Minimal issues were observed in section retrieval due to explicit SQL queries.
- No critical failure cases were reported during controlled testing.

7.7 Comparative Evaluation

To evaluate architectural improvements, a baseline system without year-aware retrieval or personalization was compared against the final system:

Table 7.3: Baseline vs Final System Performance

Metric	**Baseline (%)**	**Final (%)**
Retrieval Accuracy	82.6	94.5
Precision	84.1	93.2
Hallucination Rate	23.7	2.3
Response Consistency	74.3	89.5

- Year-aware retrieval improved accuracy by 11.9 percentage points.

- Hallucination was reduced by over 21 percentage points through source attribution and grounding.
- Deterministic mode routing (dropdown) contributed to consistency improvements.

7.8 Summary of Findings

- The chatbot achieved high accuracy in both catalog and real-time section modes.
- Year-aware retrieval and personalized prompting significantly improved advising precision.
- Dropdown-based mode selection simplified system design and reduced error rates.
- The hallucination rate was minimized to 2.3%, demonstrating strong grounding in retrieved content.
- Performance remained consistent across multiple academic programs, validating the system's generalizability.

CHAPTER 8
Discussion and Analysis

8.1 Overview

This chapter interprets the results presented in Chapter 7 and connects them to the design choices made during system development. While raw performance metrics quantify system behavior, qualitative analysis reveals **why** the system performs as it does and what this implies for real-world academic advising contexts. The discussion focuses on four major dimensions:

- Accuracy and stability of retrieval across different academic programs.
- Effectiveness of year-aware retrieval and personalization mechanisms.
- Practical impact of dropdown-based mode selection.
- Limitations, scalability considerations, and implications for future advising systems.

8.2 Impact of Year-Aware Retrieval

One of the most significant factors contributing to system performance is the integration of year-aware retrieval. Unlike traditional advising chatbots that rely on static datasets or generic retrieval methods, this system dynamically maps each student to their correct catalog year before querying the vector database. This ensures that:

- Policies and prerequisites reflect the curriculum applicable to the individual student.

- Cross-year discrepancies (e.g., course renaming or prerequisite changes) are handled seamlessly.
- Grounding of LLM responses is anchored in the correct historical context.

The evaluation results showed an 11.9% increase in retrieval accuracy compared to a non-year-aware baseline. This aligns well with expected outcomes in advising systems, where catalog year mismatches are a major source of student confusion. In practice, this feature can reduce manual clarification workload for academic advisors.

8.3 Personalization and Contextual Relevance

The addition of personalized student profiles significantly increased the relevance of chatbot responses, particularly for queries that required understanding of academic progress. This was evident in advising scenarios such as:

- "What courses should I take next semester?"
- "Can I register for CS 310?"
- "Have I met the prerequisites for the advanced AI course?"

By injecting the student's completed and in-progress courses, major, and catalog year into the prompt, the chatbot was able to:

- Return only applicable courses and pathways.
- Filter out irrelevant catalog content.
- Provide advice similar to that of a human advisor.

This resulted in an increase of evaluator usefulness scores from 3.4 to 4.6 out of 5. This demonstrates that personalization is not merely an enhancement but a critical component for effective academic advising systems.

8.4 Dropdown Mode Selection vs. Intent Classification

Many RAG-based chatbot systems use automatic intent classification to decide which retrieval pipeline to use. This approach, while flexible, often introduces ambiguity and misrouting errors. In contrast, the implementation of a **user-facing dropdown menu** for mode selection in this system provided:

- Deterministic routing of queries.
- Reduced error rates in mode selection.
- Simpler backend logic and fewer failure points.

The evaluation confirmed this advantage: no mode misclassification errors were recorded. This approach is particularly suitable for academic environments, where clarity and reliability often outweigh automation in critical student-facing services.

8.5 Hallucination Reduction and Response Stability

LLMs are known to hallucinate—producing content that is linguistically convincing but factually incorrect. Through explicit source attribution, year-aware retrieval, and structured context construction, the hallucination rate was reduced from 23.7% (baseline) to 2.3%. This is a substantial reduction and indicates that:

- Grounding responses in retrieved content is effective in the academic domain.
- Vector-based retrieval combined with explicit personalization provides stable input contexts for LLMs.
- Consistency checks and prompt engineering play a major role in ensuring factuality.

Such low hallucination rates are critical in academic advising, where inaccurate information can affect students' course selections, program completion timelines, and graduation eligibility.

8.6 System Responsiveness and User Experience

The chatbot achieved a mean latency of:

- 1.9 seconds for catalog queries.
- 0.8 seconds for real-time section queries.

These times are well below the acceptable thresholds for conversational user interfaces in higher education contexts, typically expected to stay under 3 seconds. Fast and stable responses support higher student engagement and trust in the system.

8.7 Error Analysis

Although overall system accuracy was high, error cases provided useful insights:

- Ambiguous prerequisite structures in certain program catalogs led to occasional retrieval overlaps.
- Some older catalog years had less structured formatting, which affected vector embedding quality.
- A few errors were traced back to incomplete onboarding records rather than technical issues.

Addressing these areas could push accuracy and stability even higher in future iterations.

8.8 Scalability and Generalization

The modular, microservice-based design enables:

- Easy addition of new catalog years or academic programs.

- Horizontal scaling with Docker and Kubernetes for increased load.
- Deployment in multi-institution environments with minimal changes to retrieval logic.

Because the retrieval pipelines are decoupled, new components (e.g., additional onboarding modules or registrar integrations) can be introduced without major redesign. This provides a scalable foundation for institutional adoption.

8.9 Comparison with Traditional Systems

Traditional advising tools often:

- Present static rule-based recommendations.
- Fail to handle changing policies or multiple catalog versions.
- Provide generic rather than personalized responses.

This system, by contrast:

- Dynamically retrieves accurate year-specific data.
- Grounds all LLM responses in authoritative sources.
- Personalizes advising pathways based on student context.

The improvement in both accuracy and reliability positions this chatbot as a more robust alternative to FAQ systems, static degree audits, or non-personalized advising bots.

8.10 Practical Implications

The implications of these results extend beyond technical metrics:

- **For students:** Faster and more reliable academic guidance, especially for routine queries.

- **For advisors:** Reduced workload from frequently asked questions, allowing focus on complex cases.
- **For institutions:** A scalable, policy-compliant advising support system with transparent logic.

The ability to accurately reflect institutional policies while remaining adaptable is particularly valuable in higher education environments that evolve yearly.

8.11 Limitations and Areas for Improvement

Despite strong performance, certain limitations remain:

- Catalog PDF quality impacts retrieval performance, especially for older archives.
- Course prerequisite structures are sometimes non-linear, requiring more advanced reasoning for edge cases.
- The current personalization layer does not yet incorporate predictive analytics (e.g., suggesting optimal course pathways).

Future iterations may include structured catalog parsers, more robust handling of complex prerequisites, and integration with degree audit systems.

8.12 Summary

The discussion underscores the effectiveness of the chosen architectural strategies:

- **Year-aware retrieval** directly improved accuracy and factual reliability.
- **Dropdown-based mode selection** simplified routing and eliminated misclassification.
- **Personalization** enhanced response relevance, creating a more human-like advising experience.

- **Hallucination control mechanisms** established trustworthiness in a critical domain.

The system not only meets its performance objectives but also establishes a scalable and dependable framework for real-world academic advising.

CHAPTER 9
Conclusion and Future Work

9.1 Conclusion

This directed research project presented the design, development, and evaluation of a **multi-modal Retrieval-Augmented Generation (RAG) chatbot for academic advising** that integrates:

- Year-aware retrieval from historical and current academic catalogs.
- Real-time course availability queries through direct database access.
- Student-specific personalization for contextual advising.
- Deterministic dropdown-based mode selection for clear query routing.
- Grounded response generation with minimized hallucination.

The motivation behind this system was to address three persistent issues in traditional academic advising:

1. Information fragmentation across multiple data sources (catalog PDFs, registration systems, onboarding workflows).
2. Temporal complexity caused by changing academic policies and multiple catalog years.
3. Lack of personalization and dynamic response generation in existing advising tools.

Through careful architectural design, a robust data ingestion pipeline, and intelligent retrieval and response strategies, the system successfully overcame these challenges. The **evaluation demonstrated an average retrieval accuracy of 94.5%**, with a hallucination rate reduced to 2.3%, and high response consistency

across academic programs. These metrics reflect both technical reliability and practical applicability in real institutional environments.

9.2 Key Contributions

This work makes several important contributions to the field of intelligent academic support systems:

- **A Year-Aware Retrieval Mechanism** that ensures factual accuracy tied to each student's catalog year, addressing a critical gap in advising automation.
- **A Multi-Modal Retrieval Pipeline** that integrates both semantic search (Qdrant) and structured SQL queries (PostgreSQL) under a unified architecture.
- **A Personalization Layer** that dynamically injects student context, improving the relevance and quality of advising responses.
- **Dropdown-Based Mode Routing** that simplifies backend logic, improves stability, and eliminates intent misclassification errors common in intent-classifier-based systems.
- **A Hallucination Prevention and Consistency Framework** that ensures chatbot responses are grounded in authoritative data.

9.3 Research Implications

This research contributes to both technical and academic advising domains:

- For the **academic domain**, it provides a scalable framework that institutions can adapt to reduce advisor workload, improve response times, and enhance student satisfaction.

- For the **NLP and RAG domain**, it demonstrates how careful retrieval and grounding strategies can significantly reduce hallucinations and improve factual accuracy in specialized domains.
- For **system design**, it highlights the power of deterministic mode selection and hybrid retrieval architectures in real-world deployments.

9.4 Limitations

Despite strong results, a few limitations remain:

- The system relies heavily on catalog PDF quality and consistency, which can vary significantly between years.
- Prerequisite chains with highly complex logic (e.g., OR-conditions across multiple programs) are not fully captured by the current embedding-based retrieval.
- While personalization enhances relevance, predictive course pathway recommendations are not yet implemented.
- Current evaluation focused on institutional data for a single university; multi-institution validation remains future work.

9.5 Future Work

This project lays a strong foundation for further research and real-world implementation. Future work will focus on:

- **Advanced Reranking and Structured Parsing:** Integrating cross-encoder reranking models and structured catalog parsers to handle complex prerequisite structures more accurately.

- **Predictive Personalization:** Leveraging student performance data and learning analytics to suggest optimal course sequences, thereby moving from reactive advising to proactive guidance.
- **Multi-Institution Deployment:** Extending the architecture to support multiple universities with different catalog formats, policies, and onboarding workflows.
- **Interactive Advising Dashboards:** Building real-time advisor interfaces to complement chatbot responses, enabling hybrid human-AI advising.
- **Longitudinal Impact Studies:** Evaluating the effect of chatbot advising on student retention, academic planning accuracy, and graduation rates.

9.6 Closing Remarks

The Academic Advising Chatbot demonstrates how emerging AI technologies, when carefully designed around real institutional structures and data, can provide meaningful and reliable academic support. Unlike generic chatbot systems, this solution prioritizes factual accuracy, contextual relevance, and user trust. With future enhancements, it has the potential to evolve into a comprehensive academic guidance ecosystem — supporting both students and advisors, improving advising efficiency, and aligning closely with institutional policies.

This research shows that **a well-grounded, personalized, and structured retrieval system can transform the academic advising landscape**, making critical information accessible, accurate, and actionable at scale.

REFERENCES

[1] A. Latham, K. Crockett, and D. McLean, "A Conversational Intelligent Tutoring System to Automatically Predict Learning Styles," *Computers & Education*, vol. 59, no. 1, pp. 95–109, 2010.

[2] P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

[3] V. Karpukhin, B. Oguz, S. Min, et al., "Dense Passage Retrieval for Open-Domain Question Answering," in *EMNLP*, 2020.

[4] W. Xiong et al., "Answering Complex Open-Domain Questions with Multi-Hop Dense Retrieval," in *ICLR*, 2021.

[5] S. Yu, X. Chen, and J. Li, "Course Recommendation System Using Machine Learning," in *International Conference on Educational Technology*, 2021.

[6] K. VanLehn, "The relative effectiveness of human tutoring, intelligent tutoring systems, and other tutoring systems," *Educational Psychologist*, vol. 46, no. 4, pp. 197–221, 2011.

[7] L. Wang, M. Zhou, and H. Wang, "Educational Chatbots: A Systematic Review," *Computers & Education*, vol. 195, p. 104729, 2023.

[8] R. Winkler and M. Soellner, "Unleashing the Potential of Chatbots in Education: A State-of-the-Art Analysis," *Academy of Management Annals*, 2020.

[9] J. Johnson et al., "Billion-Scale Similarity Search with GPUs," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2019.